Dr.Varodom Toochinda
Dept. of Mechanical Engineering
Kasetsart University

Lecture 5 Supplement 1

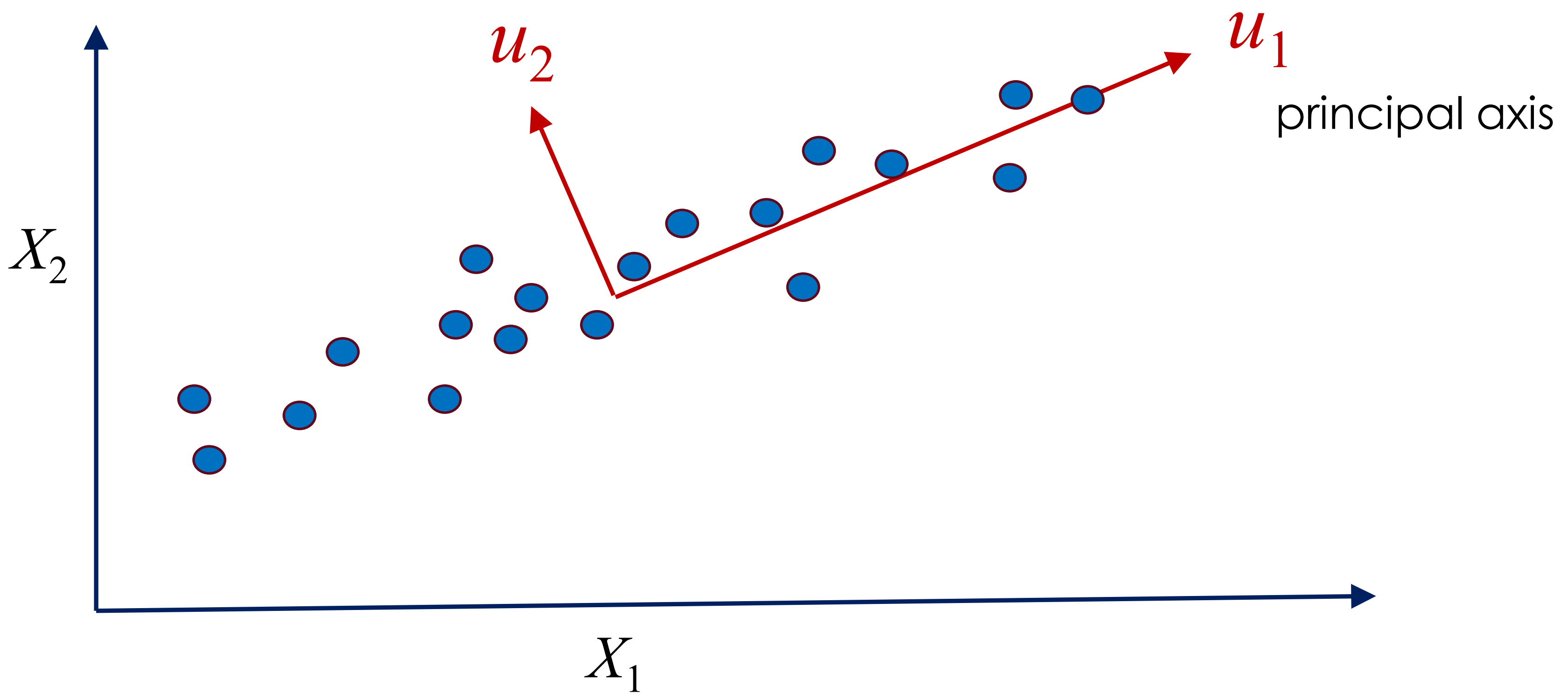
Feature Dimensionality Reduction

01211373 Machine Learning and Programming for Industry

OUTLINE

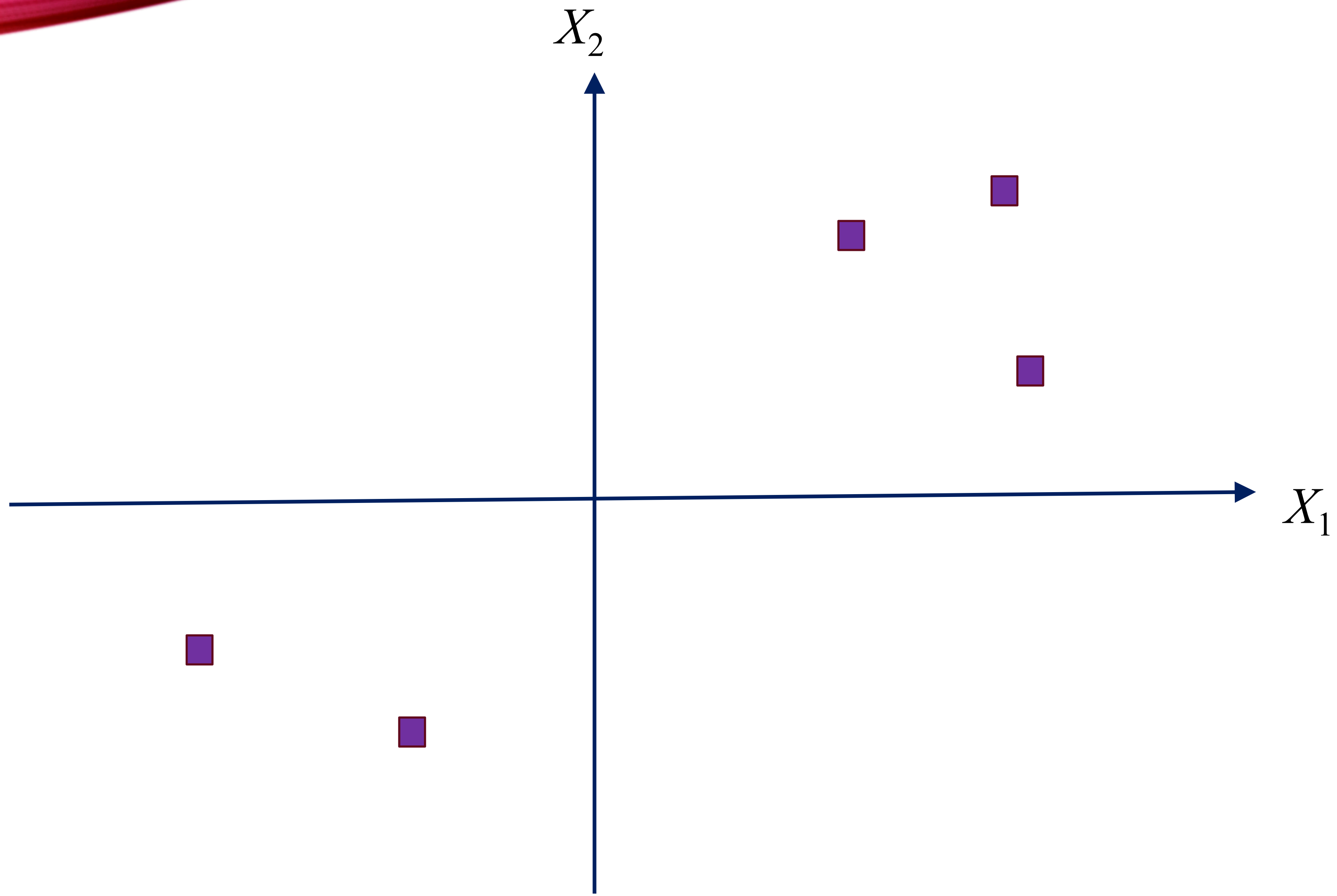
- Unsupervised
  - Principal Component Analysis (PCA)
- Supervised
  - Linear Discriminant Analysis (LDA)

PCA

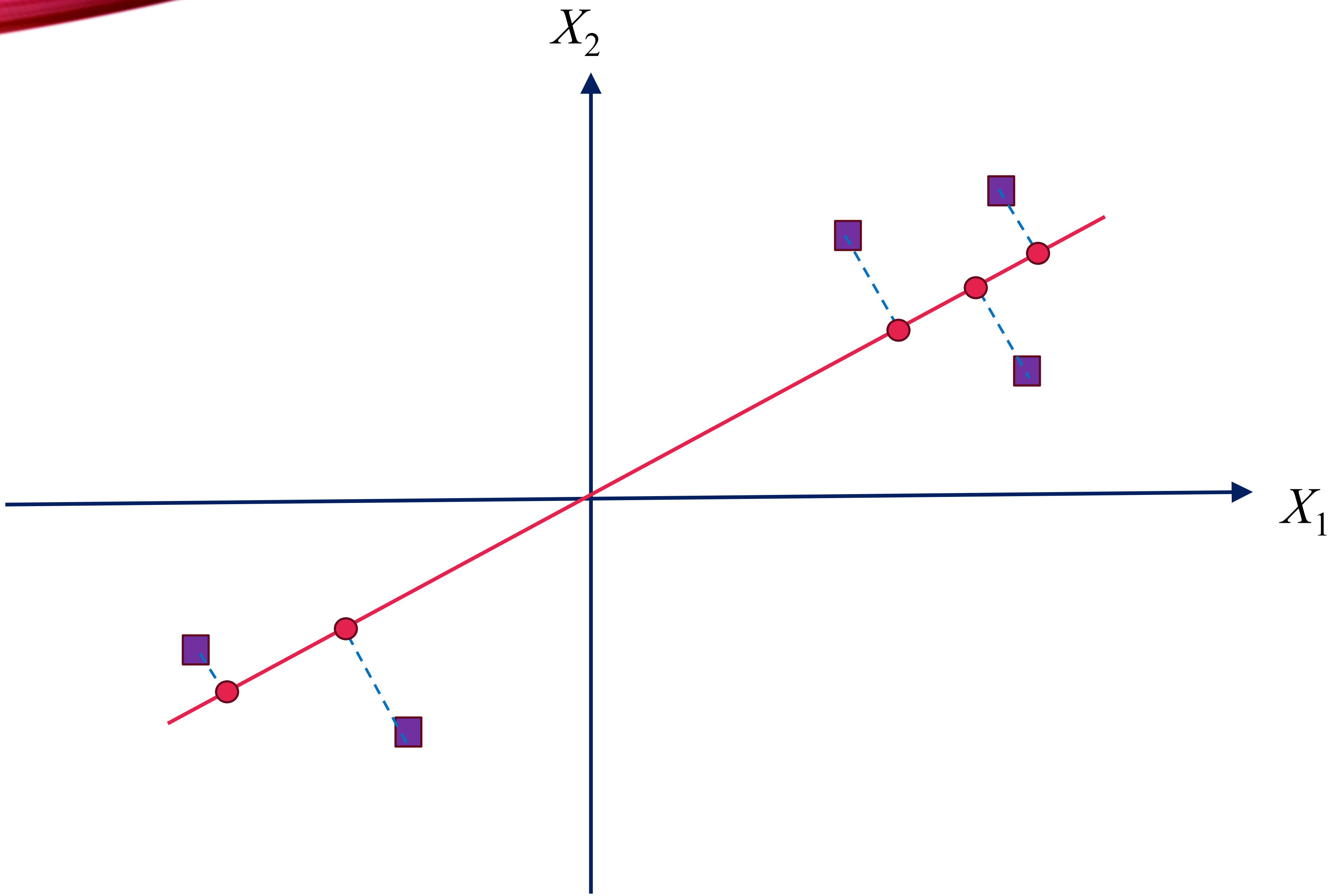


We would like to choose a direction u such that if we were to approximate the data as lying in the direction/subspace corresponding to u, as much as possible of this variance is still retained

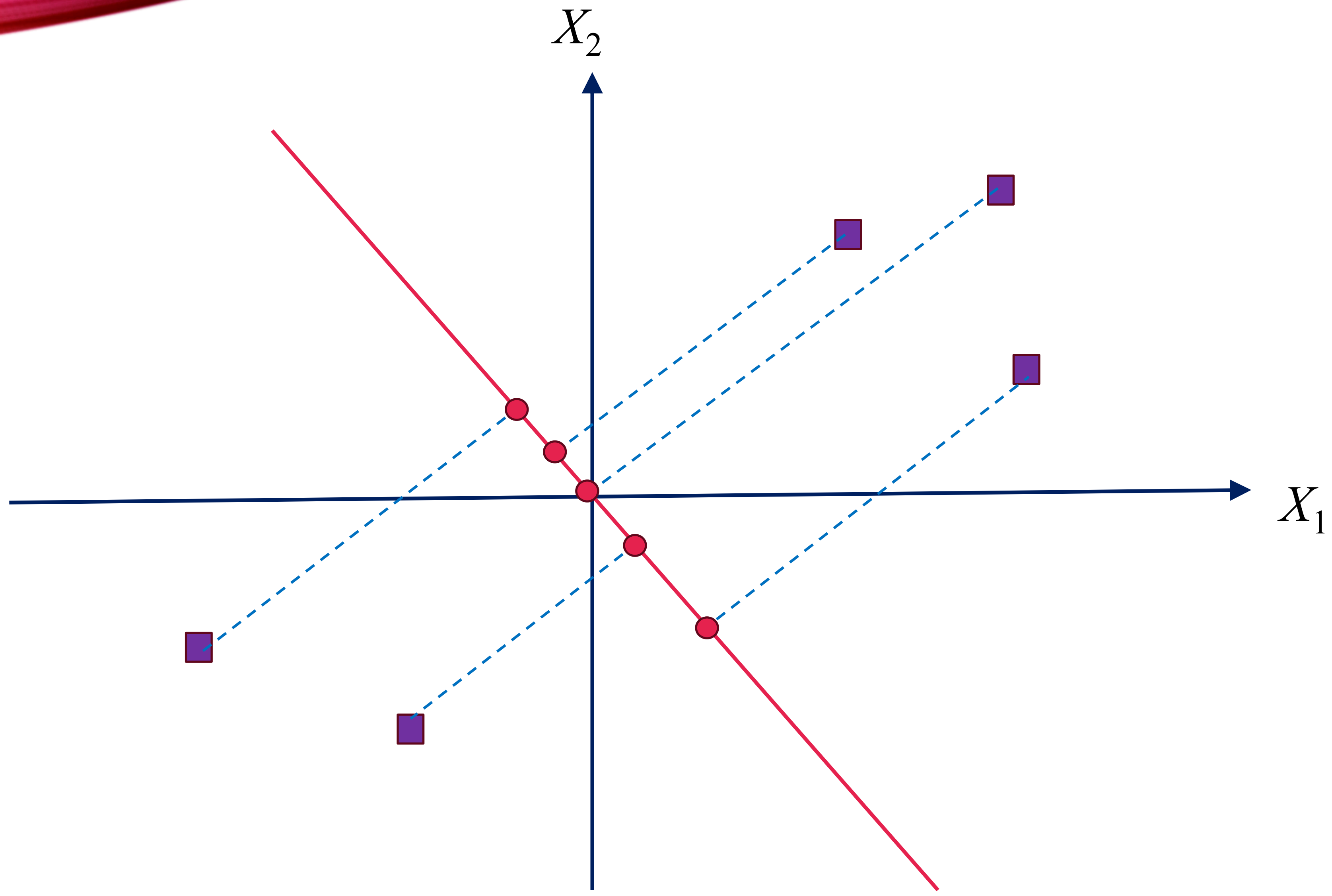
PCA



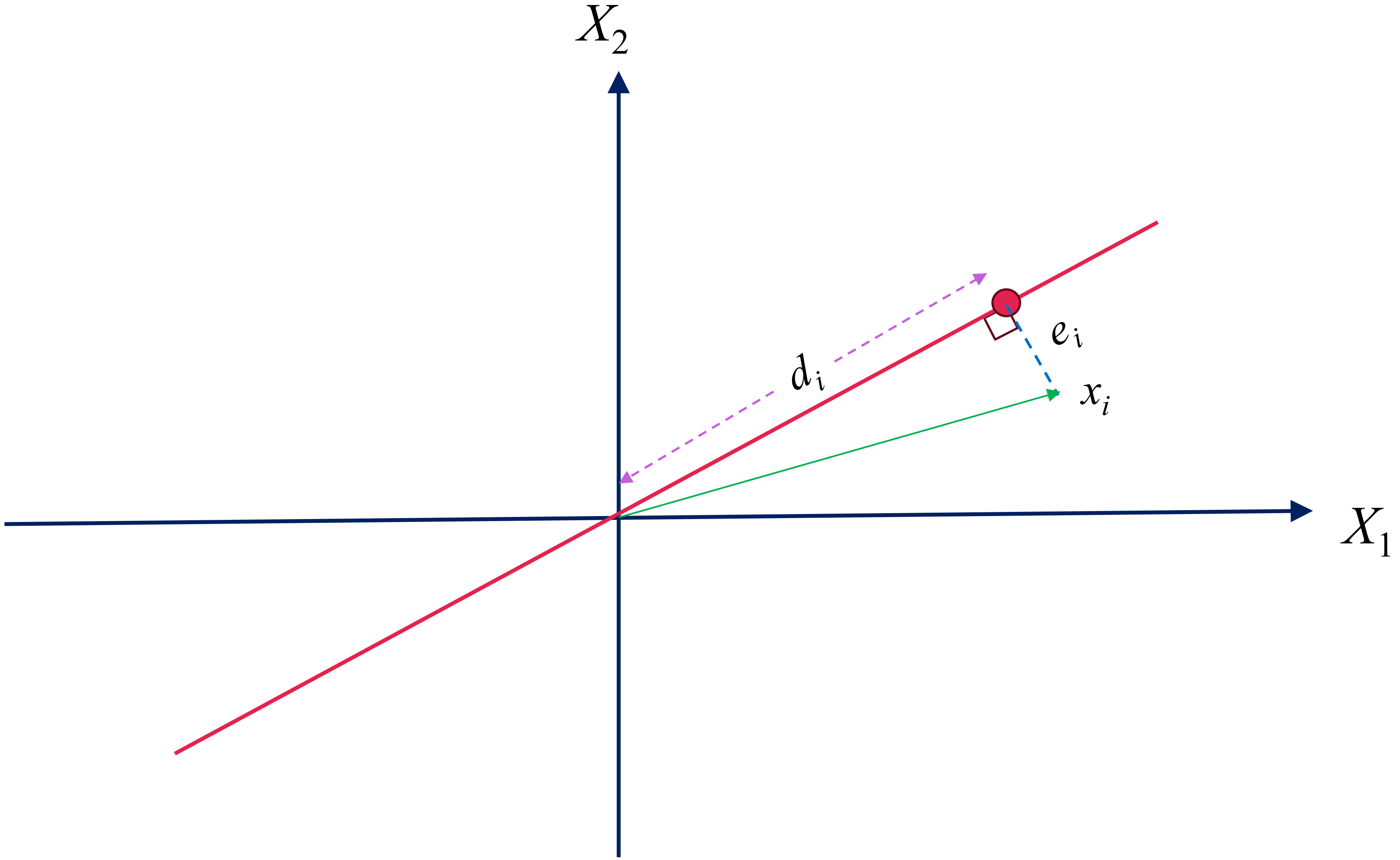
PCA



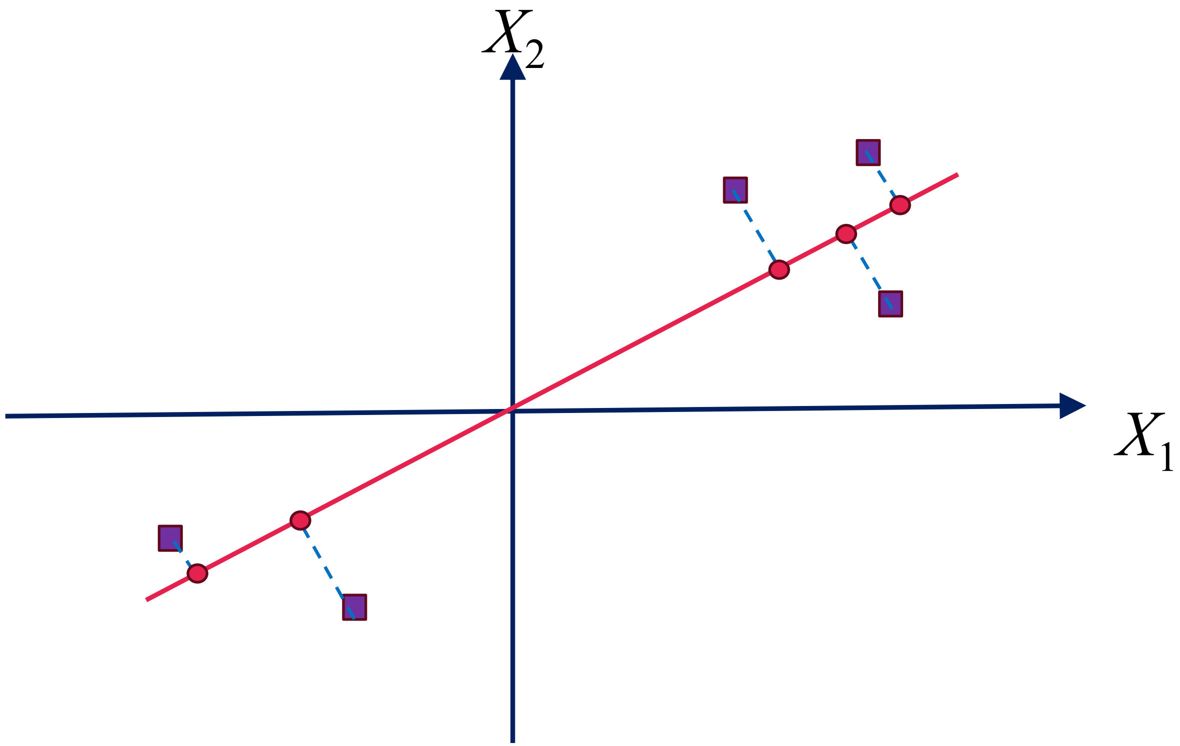
PCA



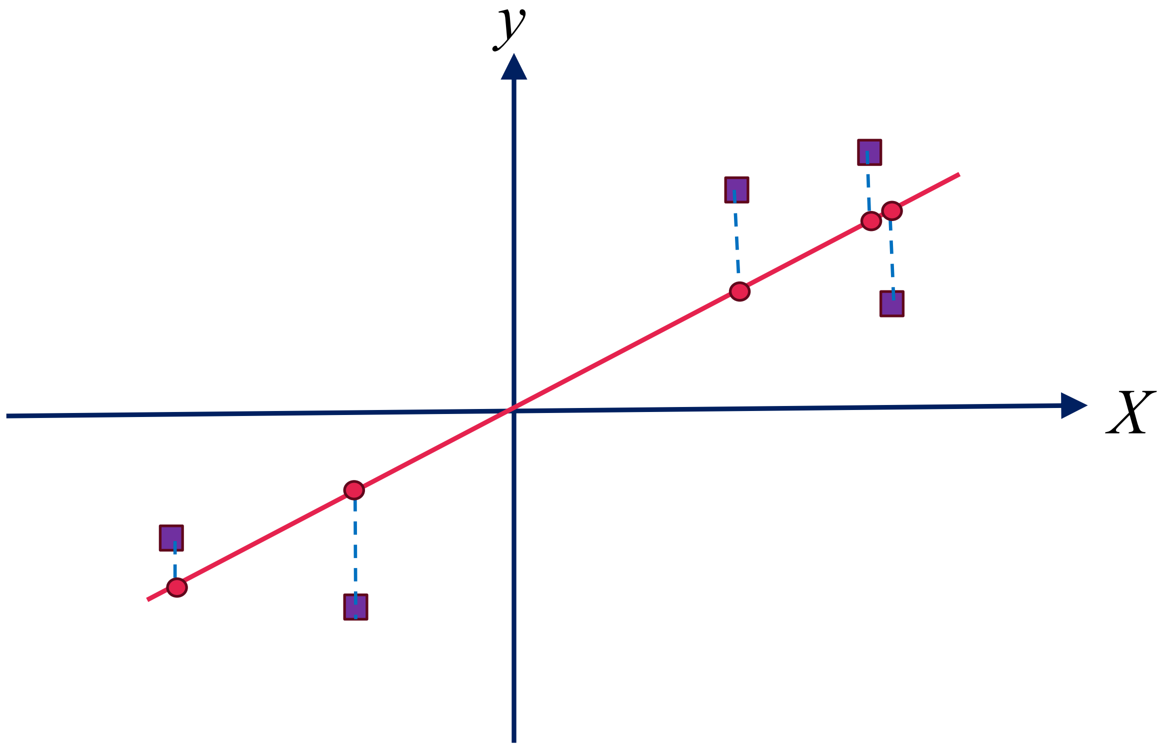
Minimizing error ⟺ maximizing variance



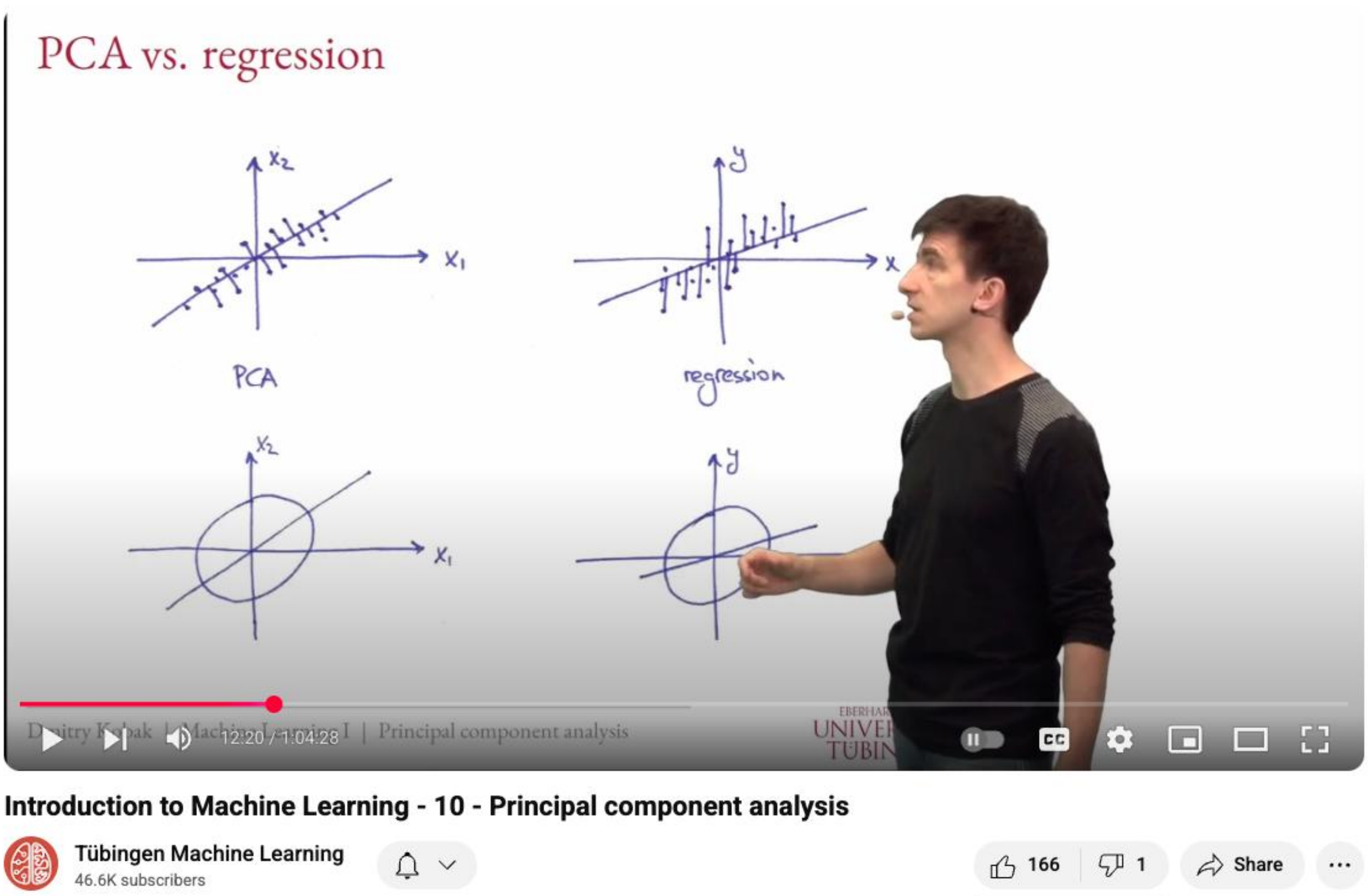
PCA



Linear regression



PCA v.s. regression



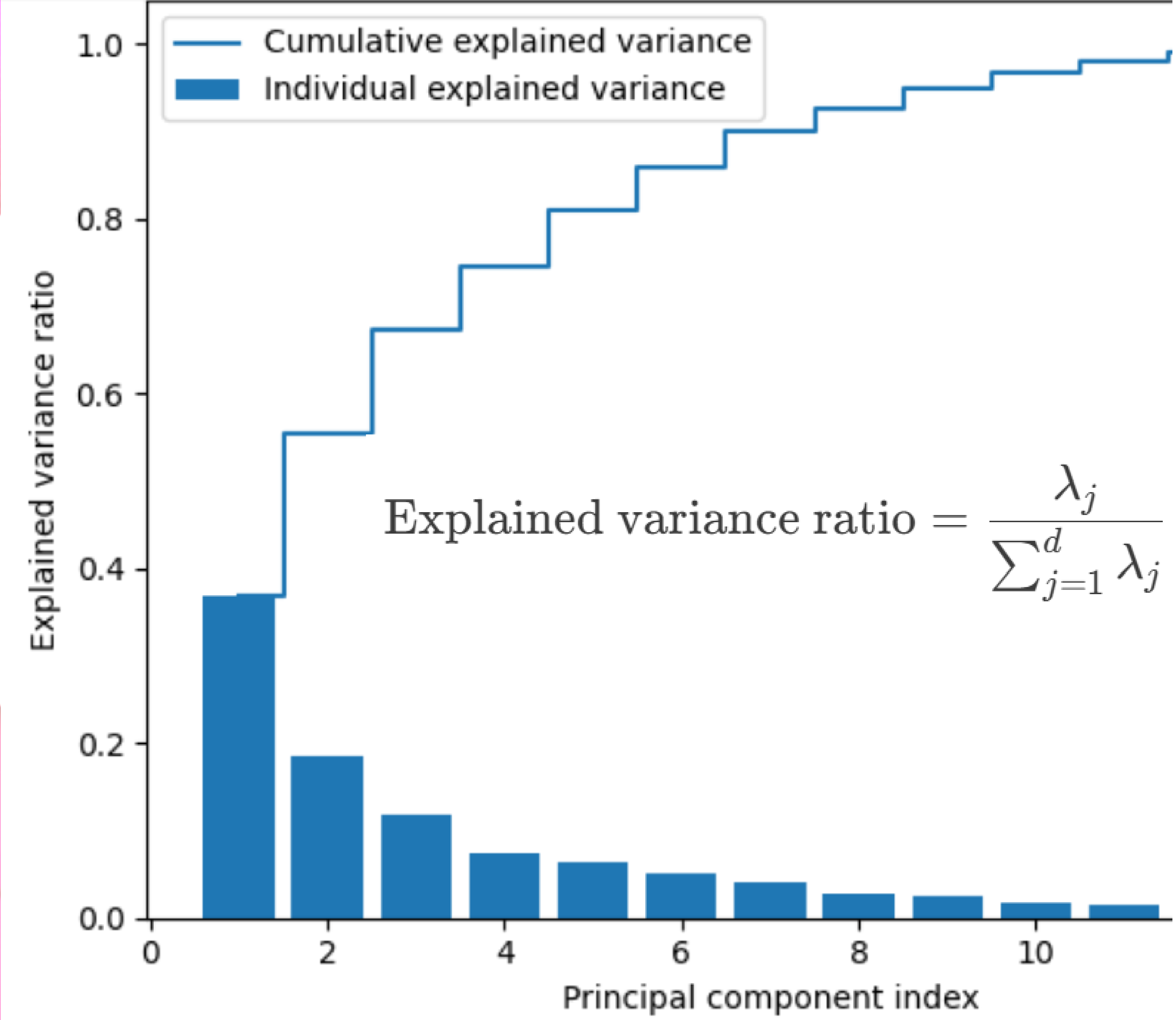
(watch this video)

https://youtu.be/xBf_LZ5ZgY4?si=PbUs_KjGuLo9BZfC&t=625

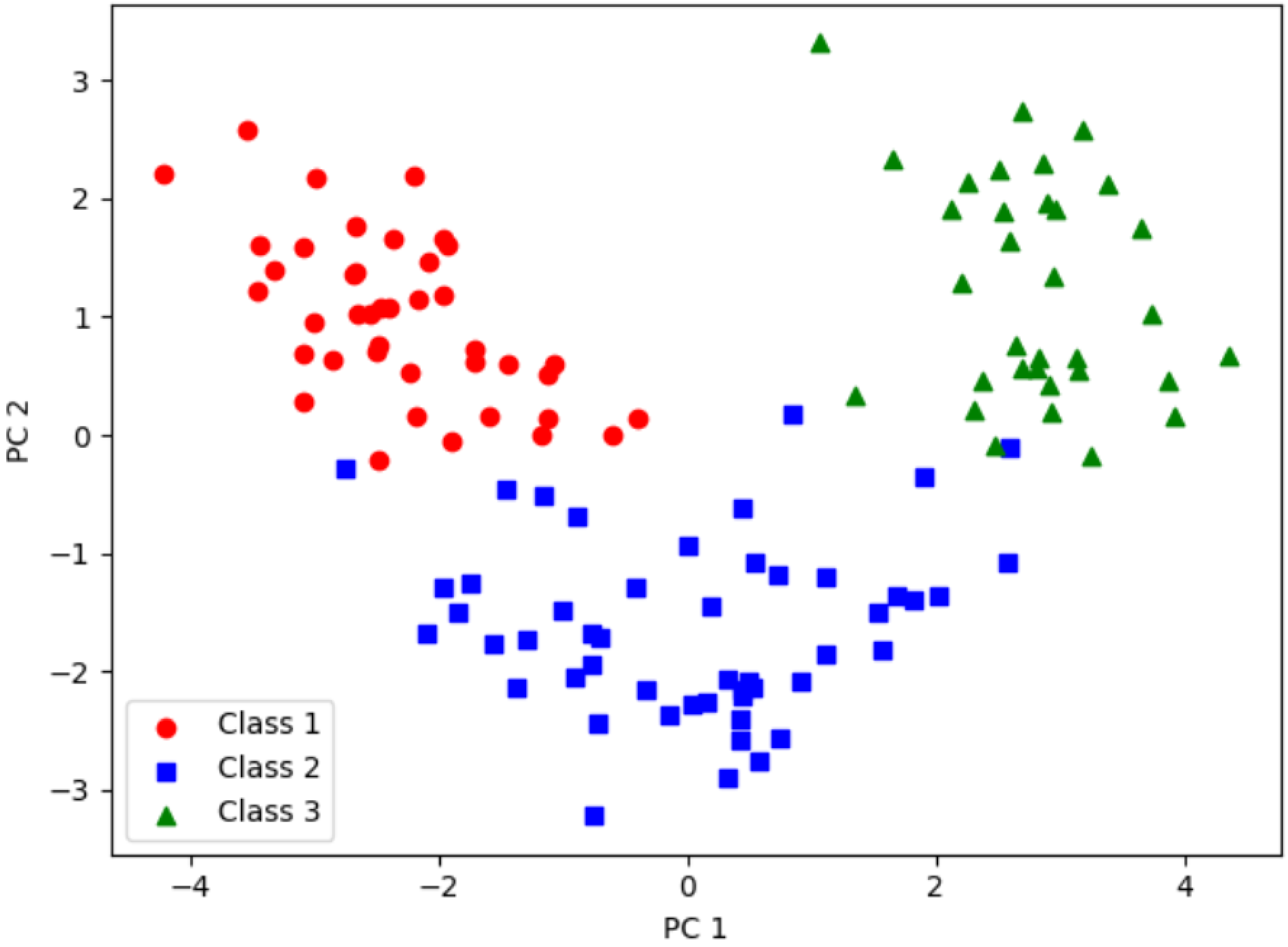
PCA with wine dataset

	Class label	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	OD280/OD315 of diluted wines	Proline
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28	2.29	5.64	1.04	3.92	1065
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735

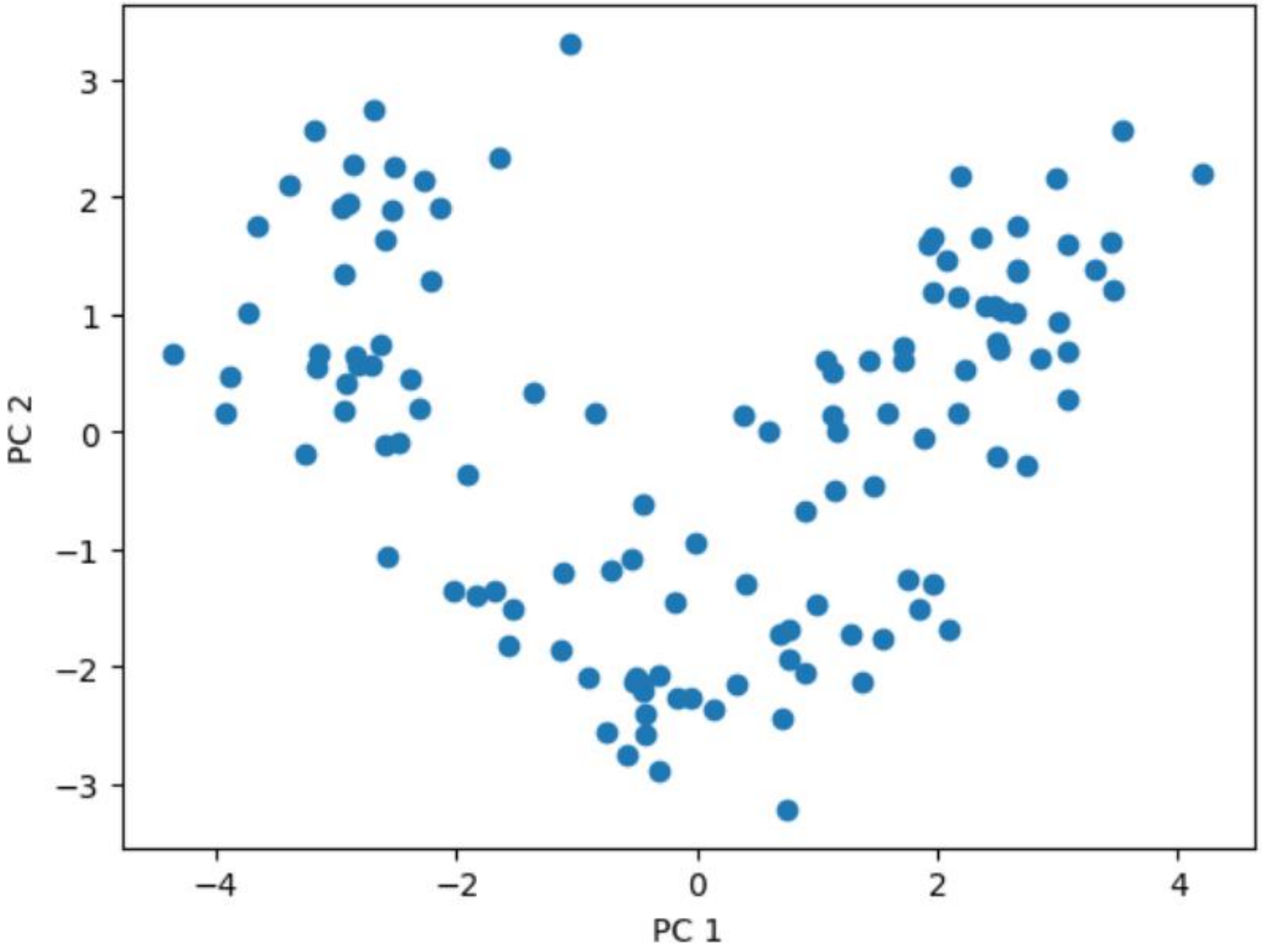
Total and explained variance



$$\text{Explained variance ratio} = \frac{\lambda_j}{\sum_{j=1}^{d} \lambda_j}$$

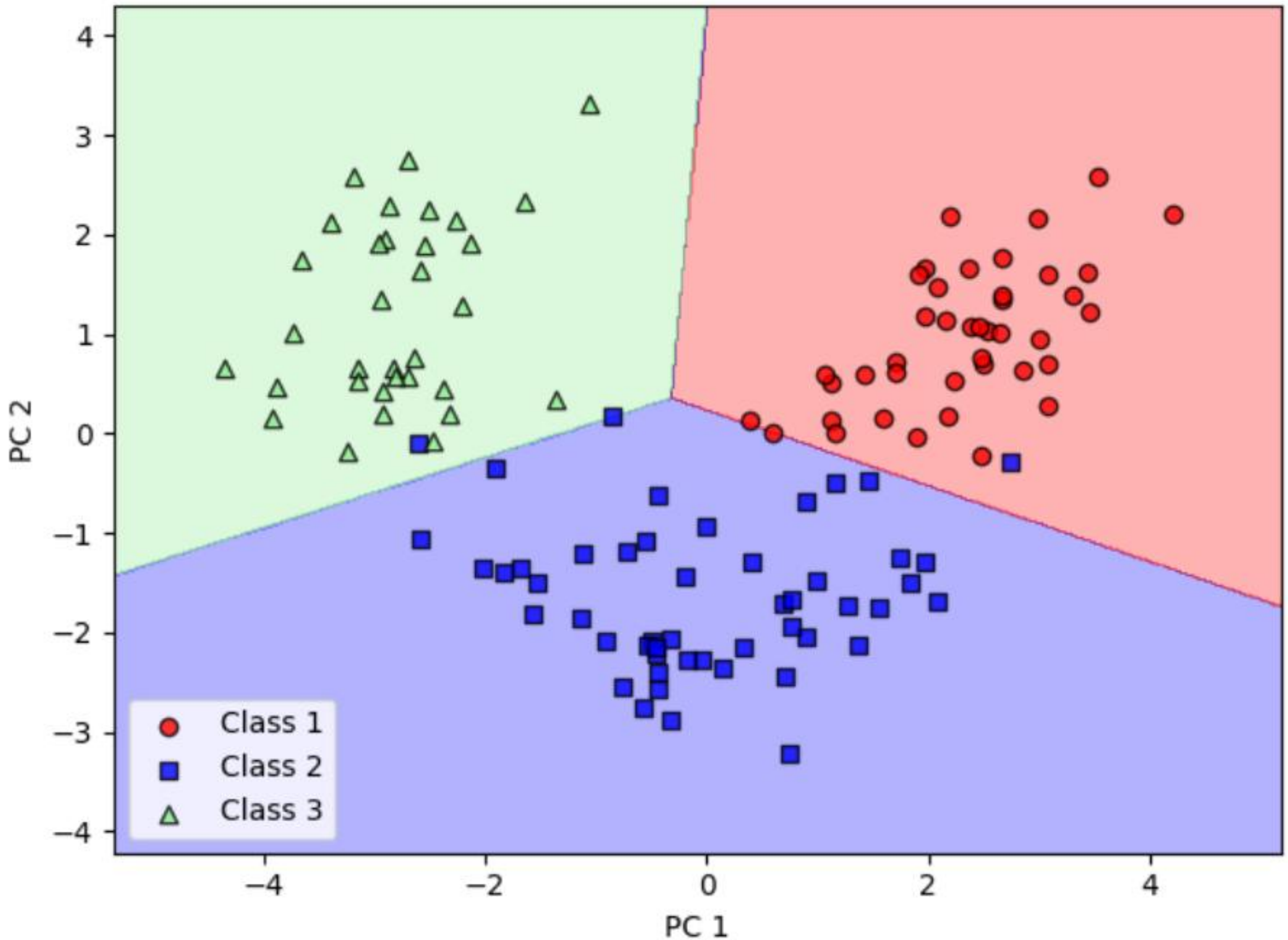


Transformed wine dataset



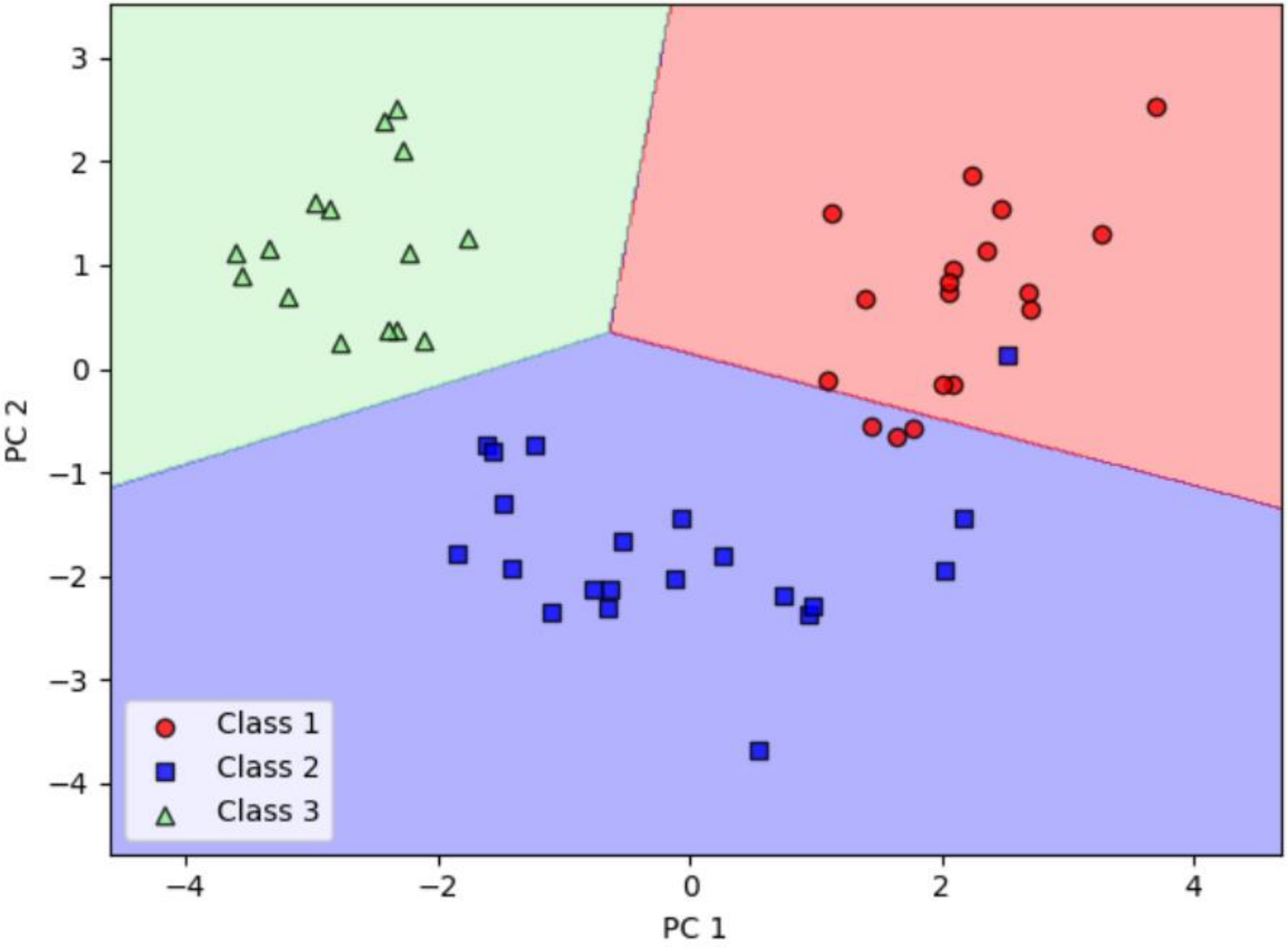
PCA in scikit-learn

(no class label information)



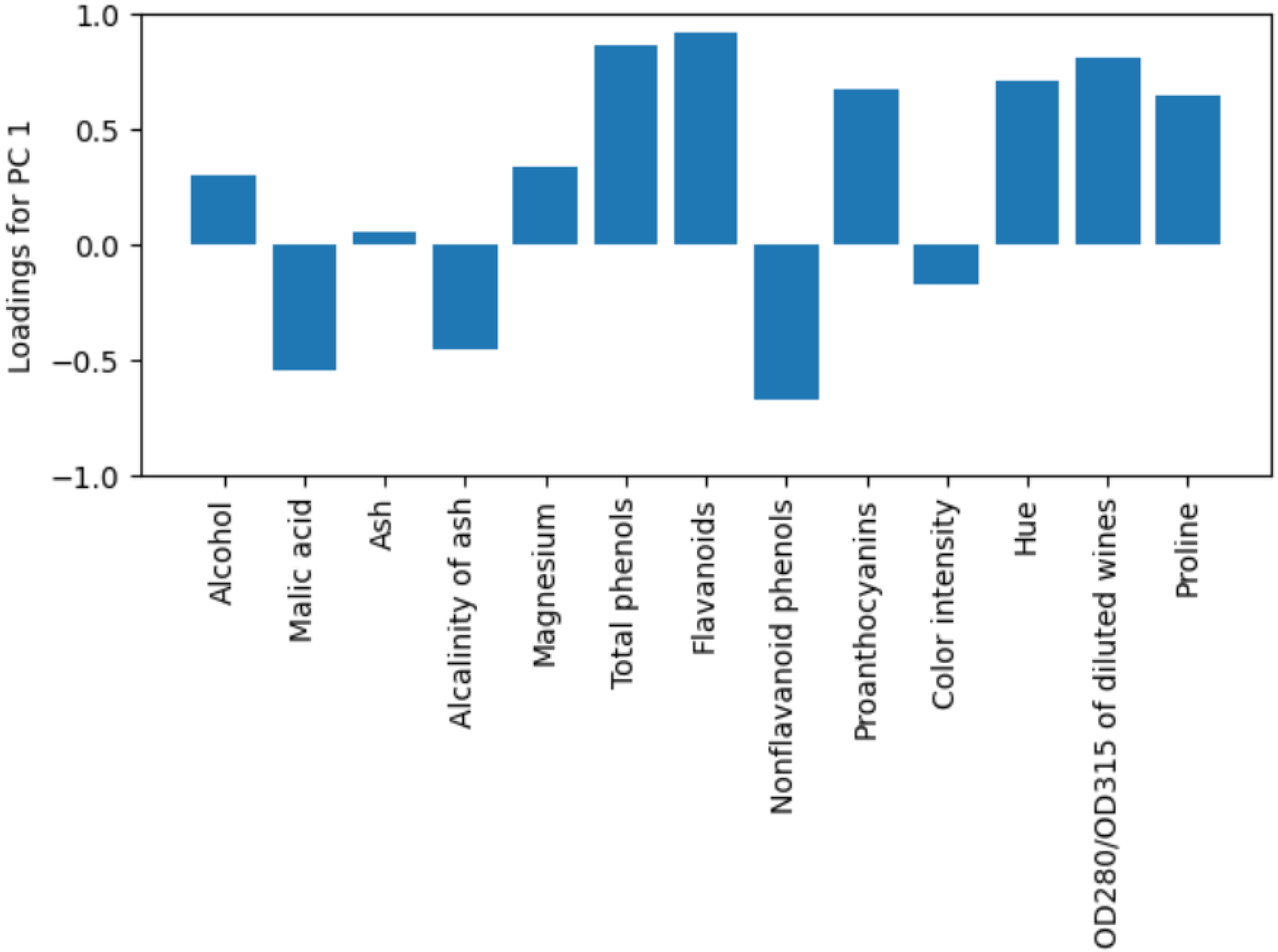
Logistic regression after PCA

(training examples)



Logistic regression after PCA

(test examples)



Assessing feature contributions

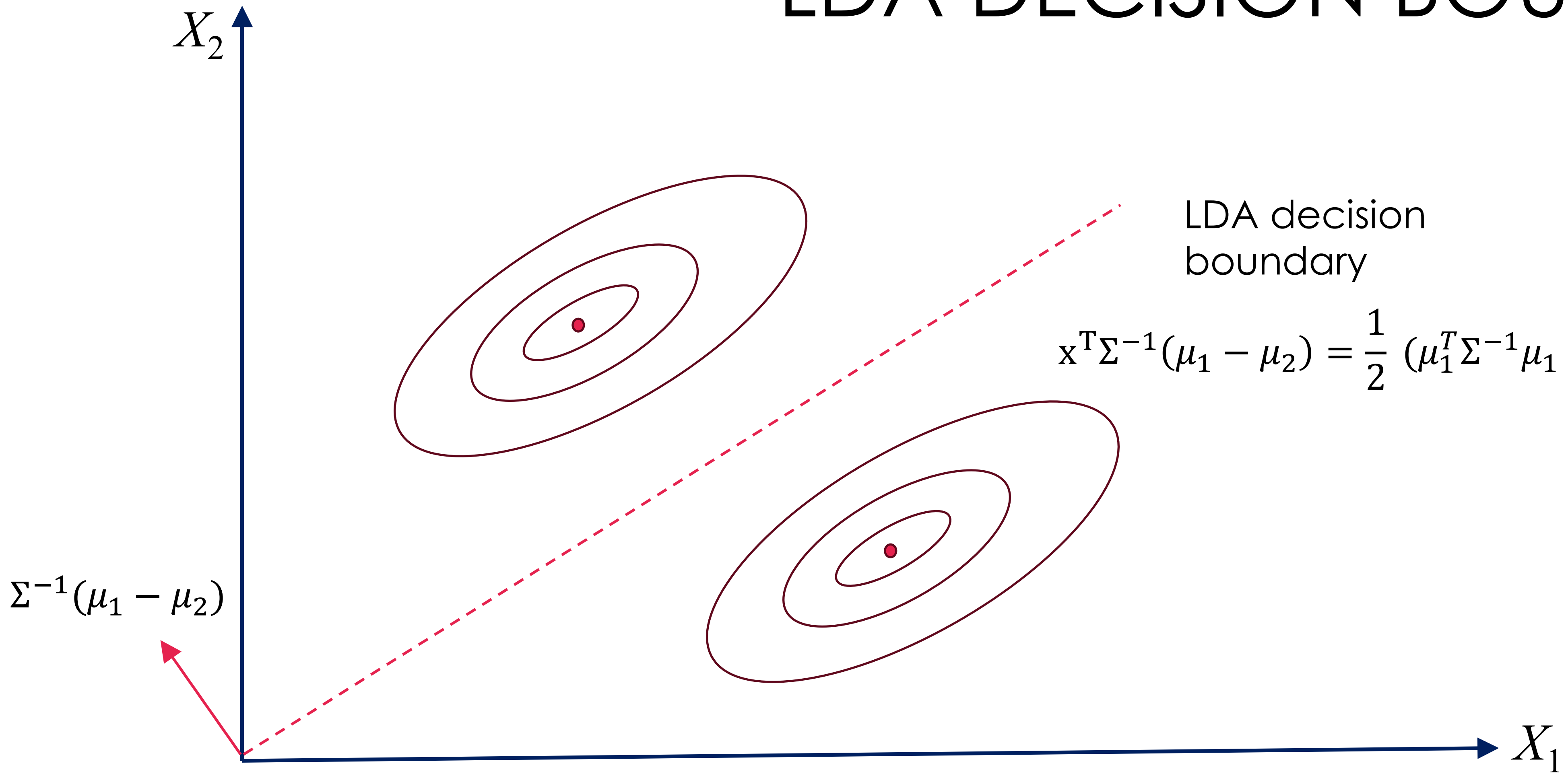
Selected Videos

- Dmitry Kobak, Introduction to Machine Learning - 10 - Principal component analysis, Tübingen Machine Learning, 2020/21
- Andrew Ng, Lecture 15 - PCA and ICA ,CS229: Machine Learning, Stanford University, 2018
- Zachary Huang, Principal Component Analysis in 30 min, 2025

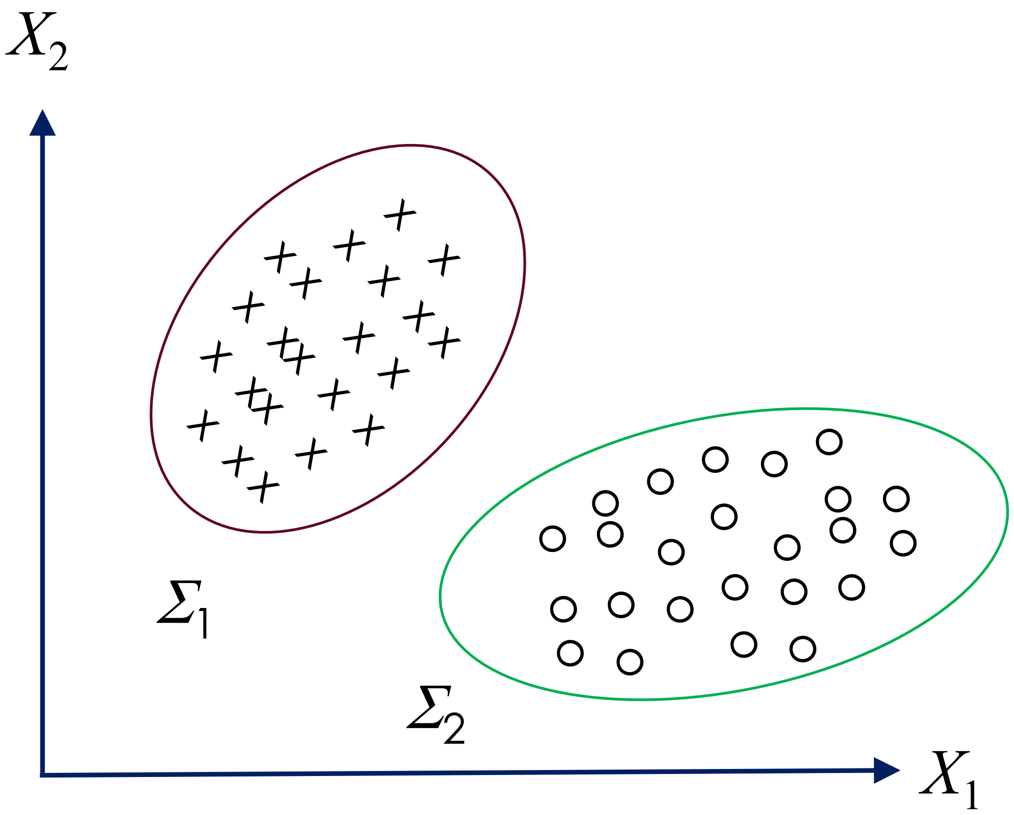
Optional

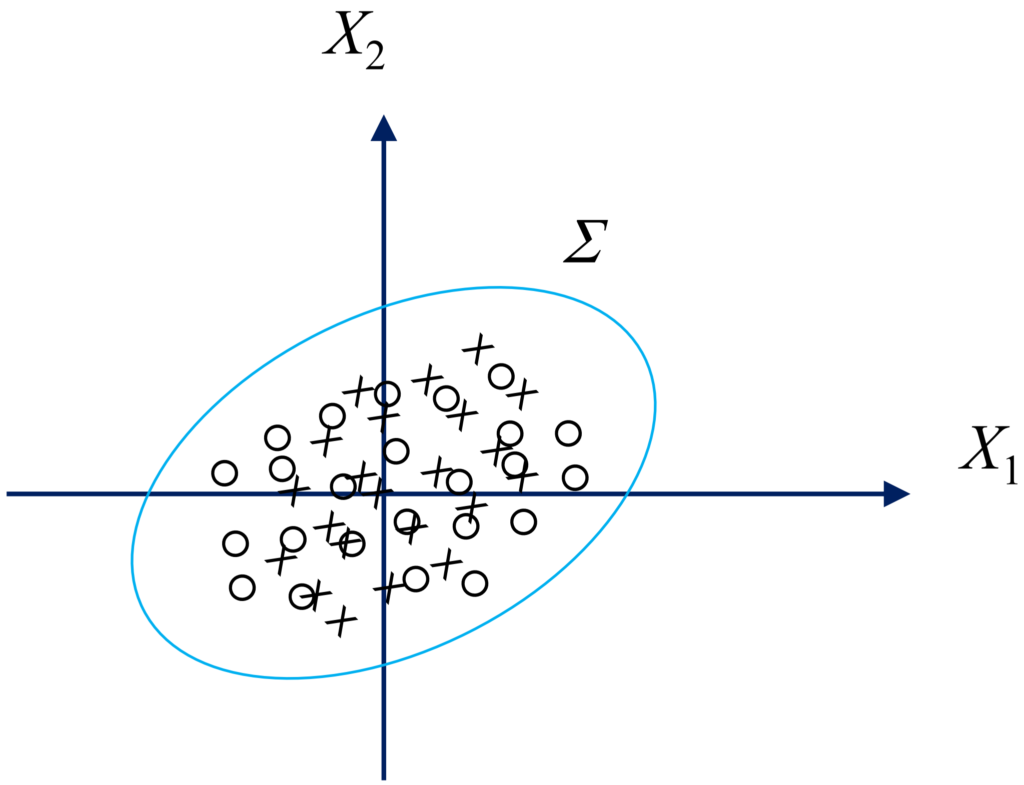
LINEAR DISCRIMINANT ANALYSIS (LDA)

LDA DECISION BOUNDARY

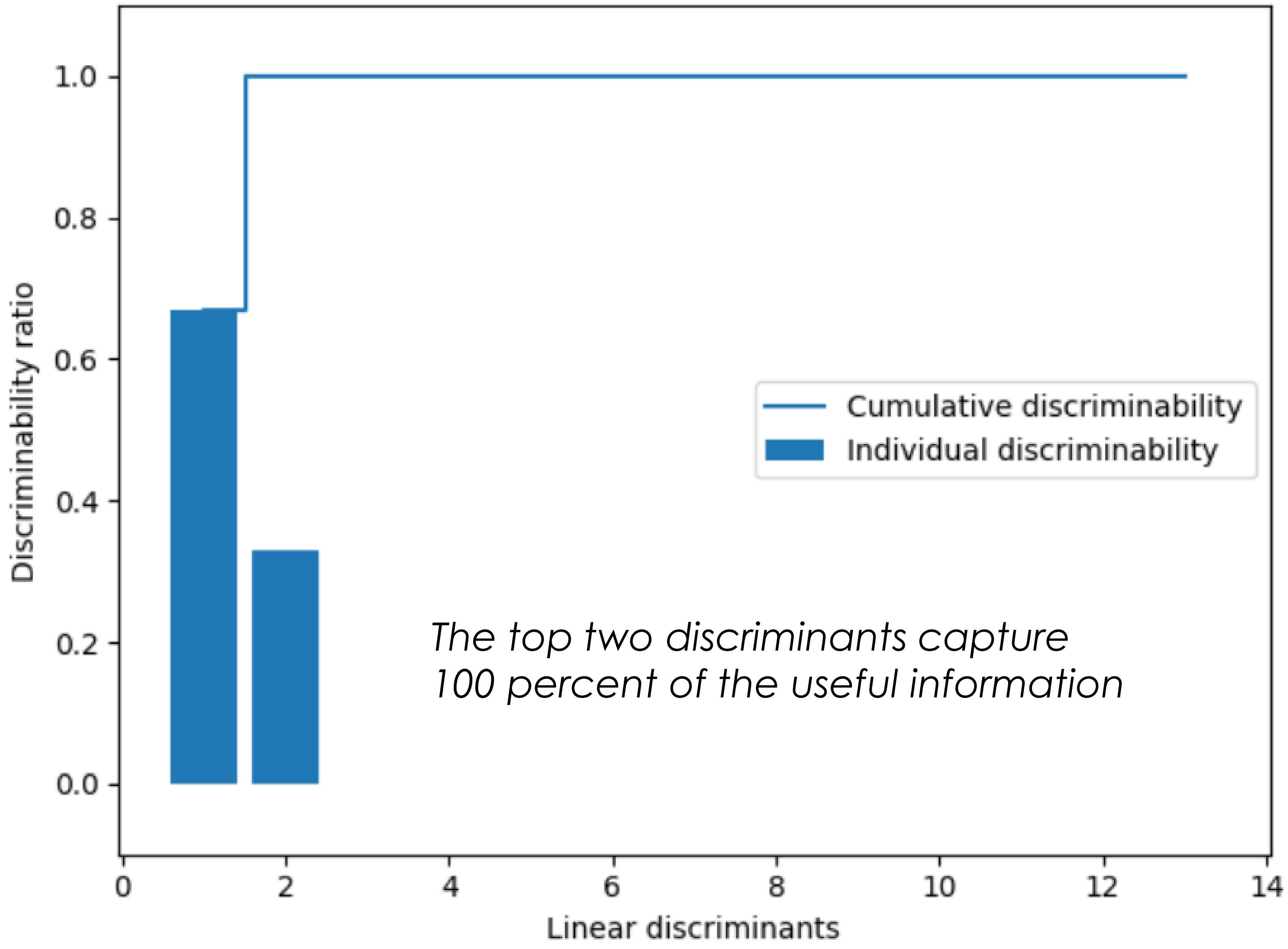


Pooled covariance

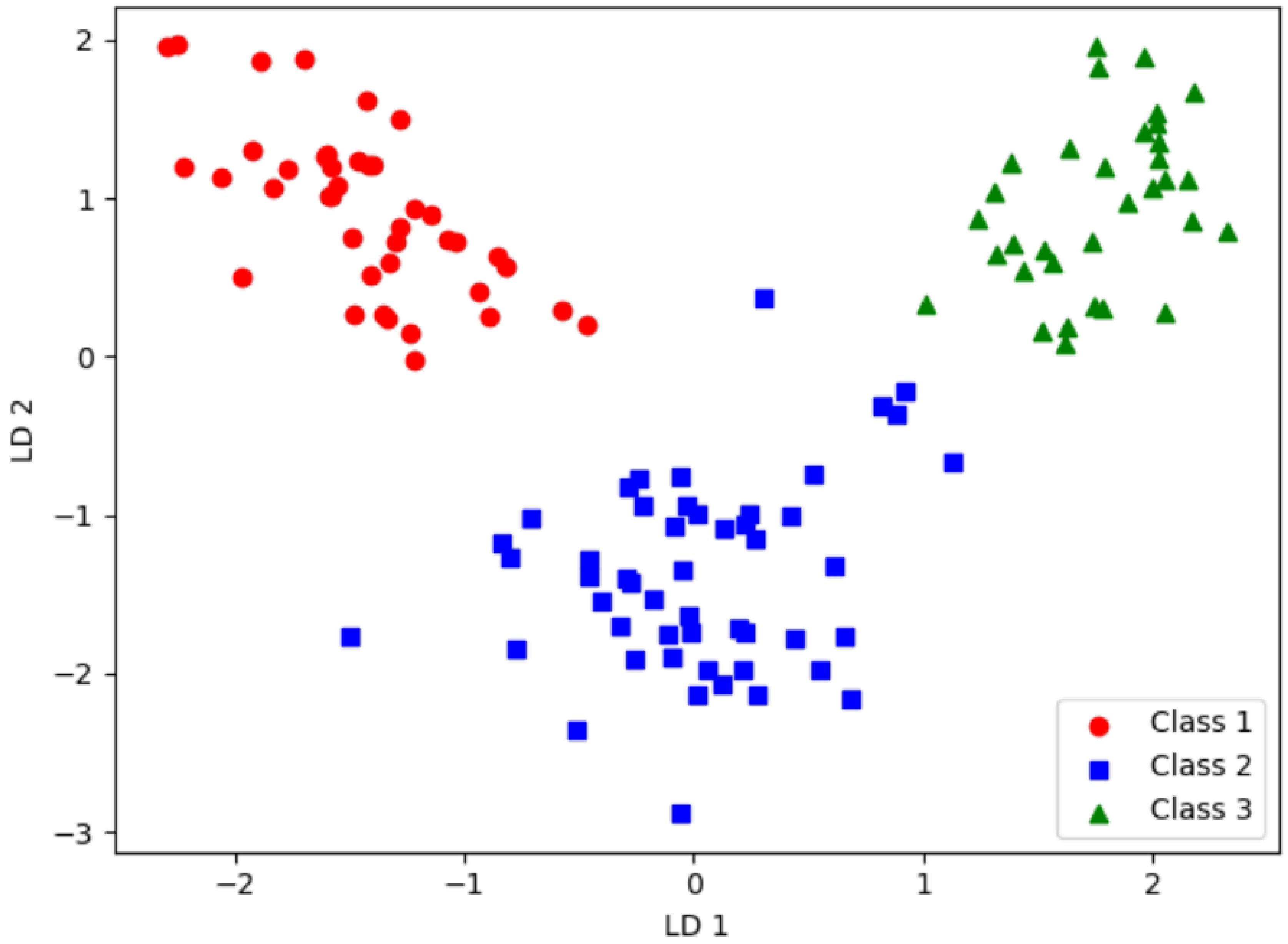


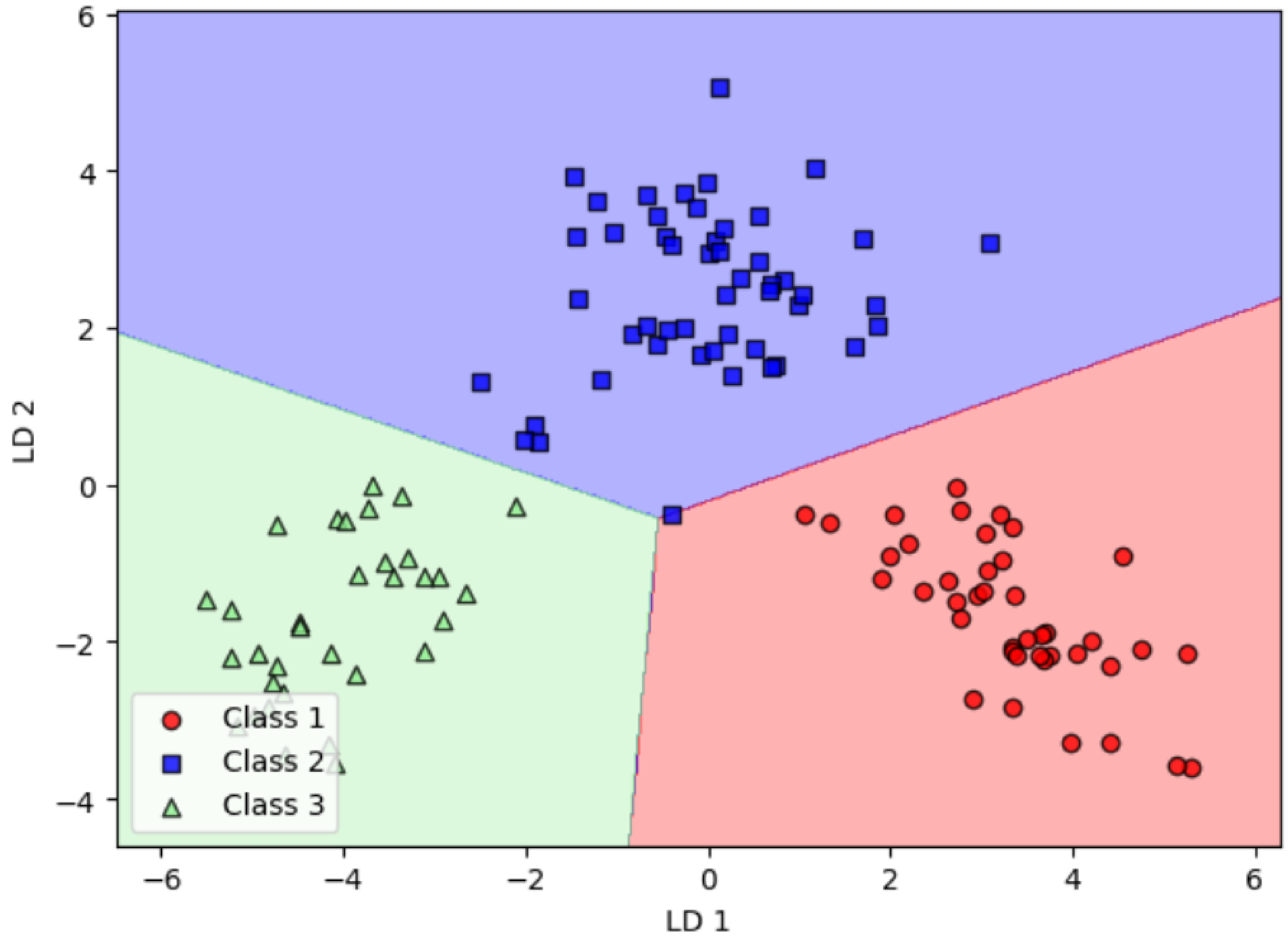


Discriminability



Projecting wine data onto first two discriminants

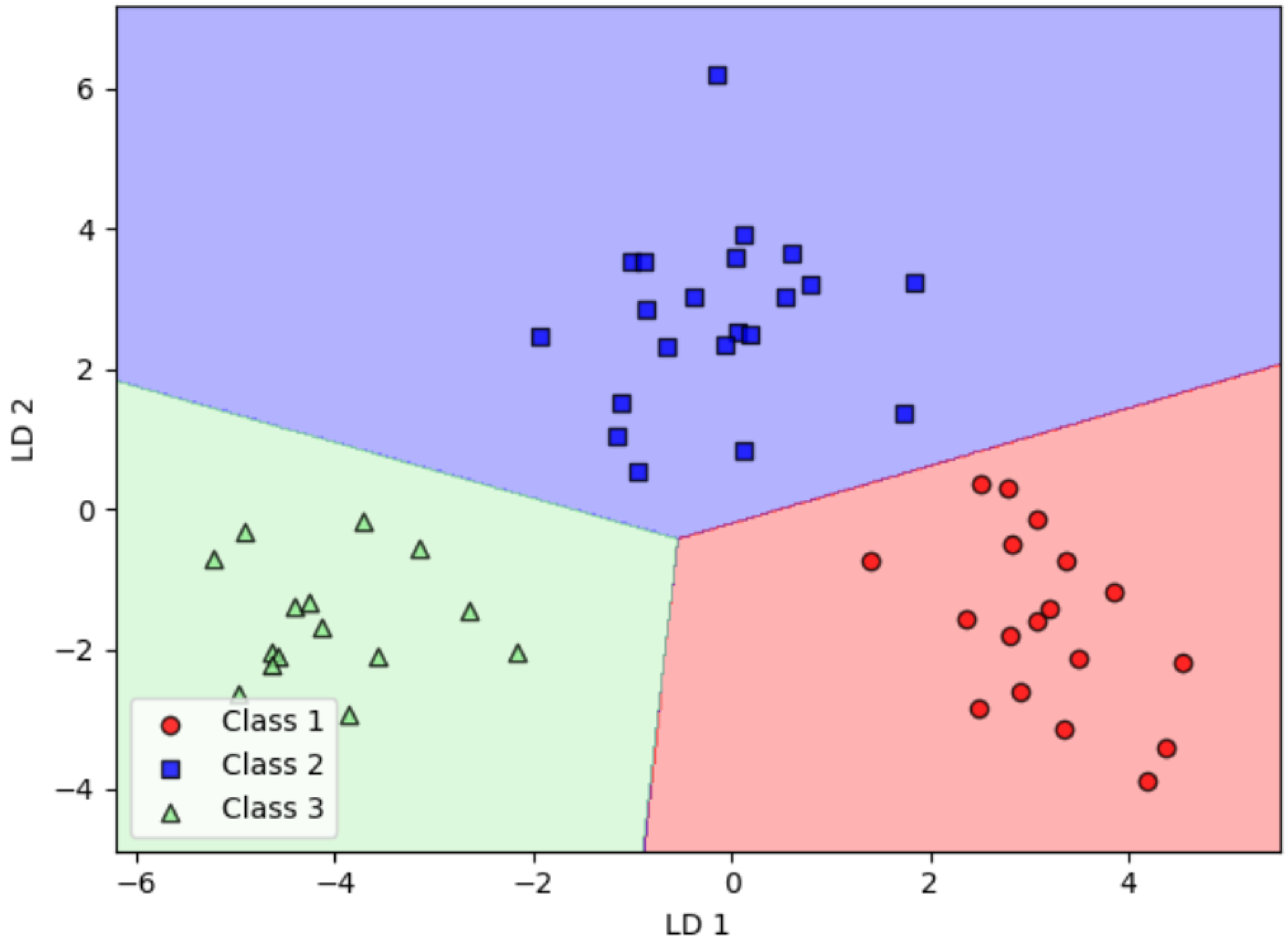




LDA via scikit-learn

Logistic regression after LDA transformation (training examples)

LDA via scikit-learn



Logistic regression after LDA transformation (test examples)

REFERENCES

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
- A. Ng. CS229 Lecture Notes. Stanford University. 2023.
- D. Kobak. Machine Learning I. University of Tübingen. 2021.